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Elimination of Left Recursion (LR)

Left and Right Recursive Grammars

In a context-free grammar G , if there is a production in the form $X \rightarrow Xa$ where X is a non-terminal and 'a' is a string of terminals, it is called a left recursive production. The grammar having a left recursive production is called a left recursive grammar.

And if in a context-free grammar G , if there is a production in the form $X \rightarrow aX$ where X is a non-terminal and 'a' is a string of terminals, it is called a right recursive production. The grammar having a right recursive production is called a right recursive grammar.

Left Recursion (LR)

$$A \longrightarrow A \alpha \mid \beta$$

Right Recursion (RR)

$$A \longrightarrow \alpha A \mid \beta$$

Left Recursion (LR)

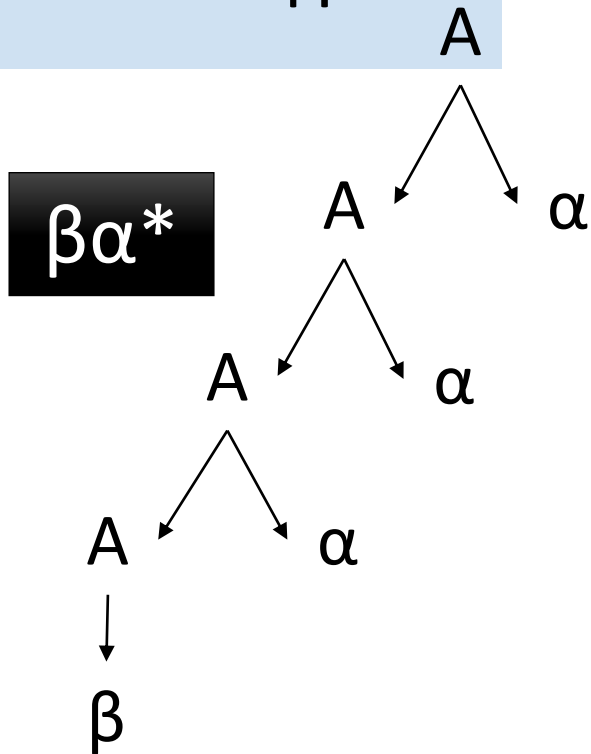
$$A \longrightarrow A \alpha \mid \beta$$

Right Recursion (RR)

$$A \longrightarrow \alpha A \mid \beta$$

Left Recursion (LR)

$$A \rightarrow A\alpha \mid \beta$$

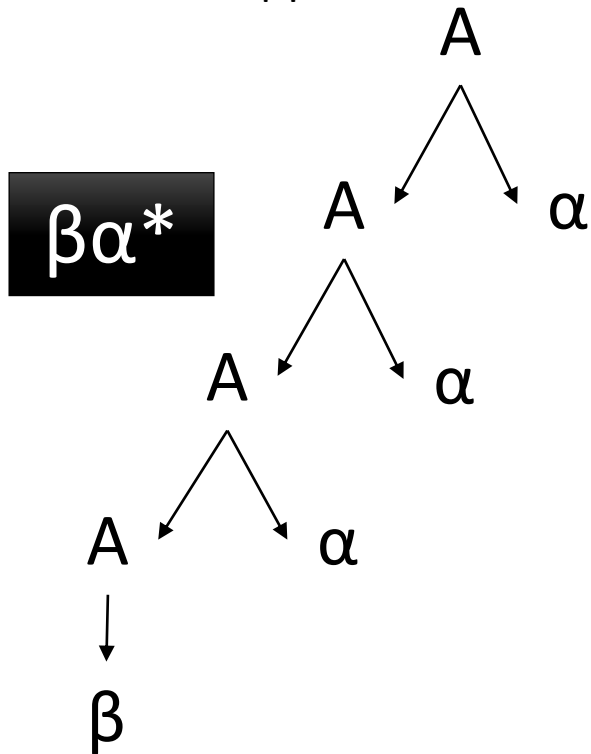


Right Recursion (RR)

$$A \rightarrow \alpha A \mid \beta$$

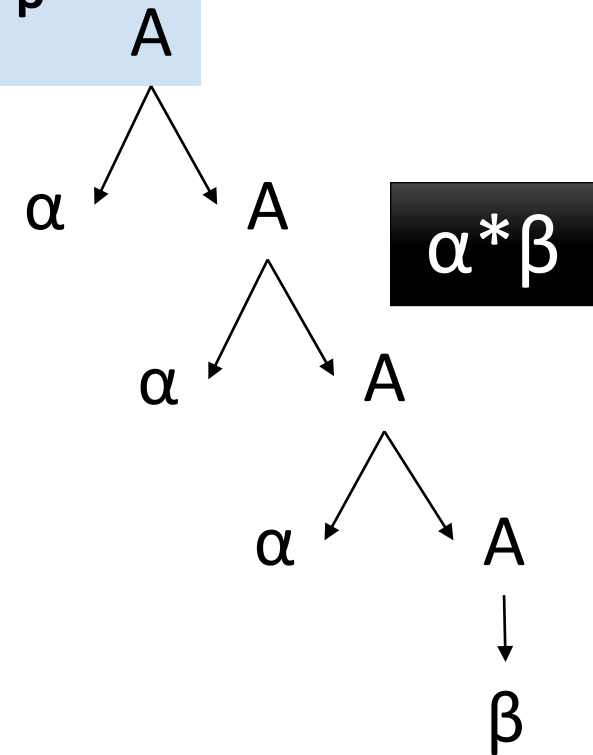
Left Recursion (LR)

$$A \longrightarrow A \alpha \mid \beta$$



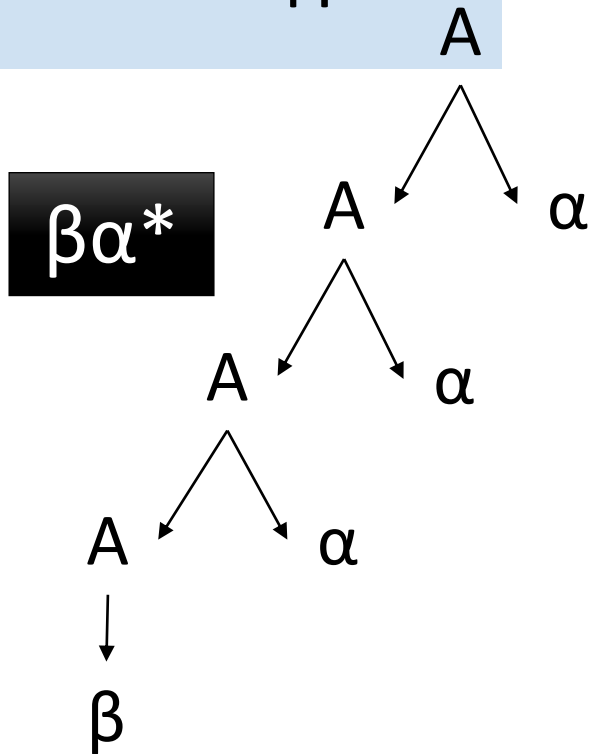
Right Recursion (RR)

$$A \longrightarrow \alpha A \mid \beta$$



Left Recursion (LR)

$$A \longrightarrow A \alpha \mid \beta$$



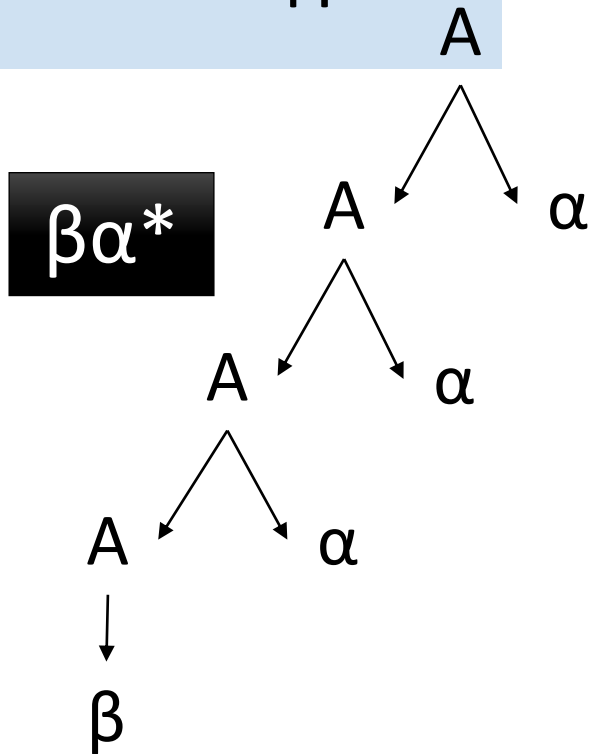
$$A \longrightarrow A \alpha \mid \beta$$

$$A \longrightarrow \beta A'$$

$$A' \longrightarrow \epsilon \mid \alpha A'$$

Left Recursion (LR)

$$A \longrightarrow A \alpha \mid \beta$$



$$A \longrightarrow A \alpha \mid \beta$$

$$A \longrightarrow \beta A'$$

$$A' \longrightarrow \epsilon \mid \alpha A'$$

Example:

$$E \longrightarrow E + T \mid T$$

Left Recursive

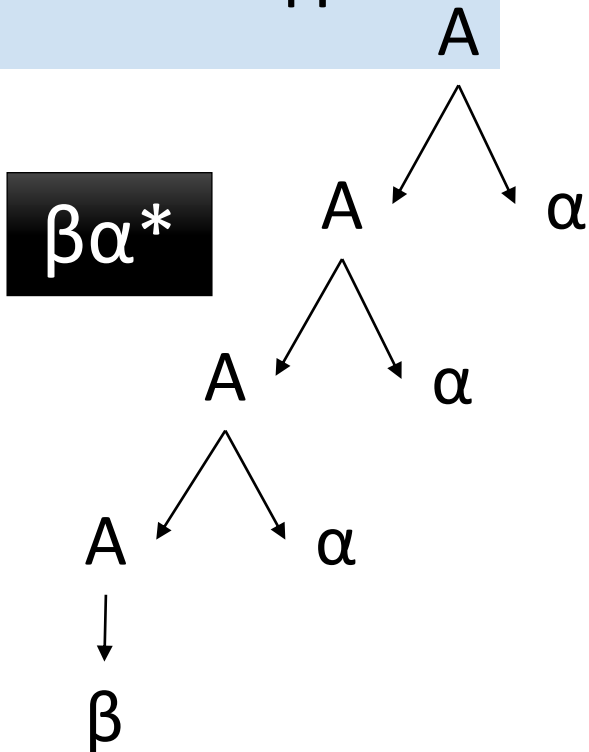
Solve:

$$E \longrightarrow T E'$$

$$E' \longrightarrow \epsilon \mid + T E'$$

Left Recursion (LR)

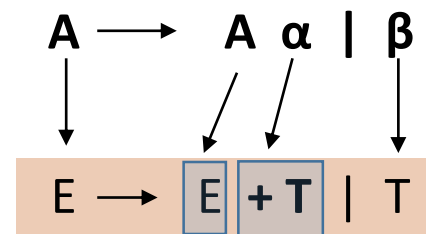
$$A \longrightarrow A \alpha \mid \beta$$



$$A \longrightarrow \beta A'$$

$$A' \longrightarrow \epsilon \mid \alpha A'$$

Example:



Solve:

$$E \longrightarrow T E'$$

$$E' \longrightarrow \epsilon \mid + T E'$$

Exercise

Exercise 1: $T \rightarrow T * F \mid F$

Exercise 2: $F \rightarrow (E) \mid \text{id}$

Exercise 3: $E \rightarrow E + T \mid T$
 $T \rightarrow T * F \mid F$
 $F \rightarrow (E) \mid \text{id}$

THANK YOU